

LEDGER · 2026-09-01 · POLYMER CHEMISTRY

Terephthalaldehyde-Crosslinked Polyimine Vitrimers via Catalyst-Free Solution Casting

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Bisphenol A (BPA) Epoxy Resin
Measured advantage	Tensile strength of 97.0 MPa vs 70-80 MPa for BPA Epoxy (1.2x improvement); 5 reprocessability cycles vs 0 for BPA Epoxy
Time to first revenue	4 months
Gross margin at parity	70.0%
Startup CapEx	\$85,000
Payback from first sale	1.4 months
Capital productivity	19.2x
Regulatory risk	Moderate — qualification costs reported (\$45,000 estimated)

Assessment

While the vitrimer achieves excellent unit economics and rapid speed to market, its tensile strength gain over BPA epoxy is only 1.2x (97 MPa vs 70-80 MPa), falling short of the required 2x step-change bar. The reprocessability advantage represents a circularity/sustainability preference attribute rather than a performance dimension the target buyer currently pays for. Consequently, the concept fails the strict superiority gate.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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Method

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